Name: _____

Class: _____

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Algorithms 2 — Flowcharts and Trace Tables

1 The six flowchart symbols

VOCAB **Flowchart** (OCR 2.1.2) — a diagram of an algorithm, drawn with standard symbols joined by arrows that show the order of the steps

A flowchart shows the same steps as a program, as a picture. OCR uses six symbols. In the exam you may be asked to read a flowchart, complete one, or draw your own, and a Section B design answer can be given as a flowchart.

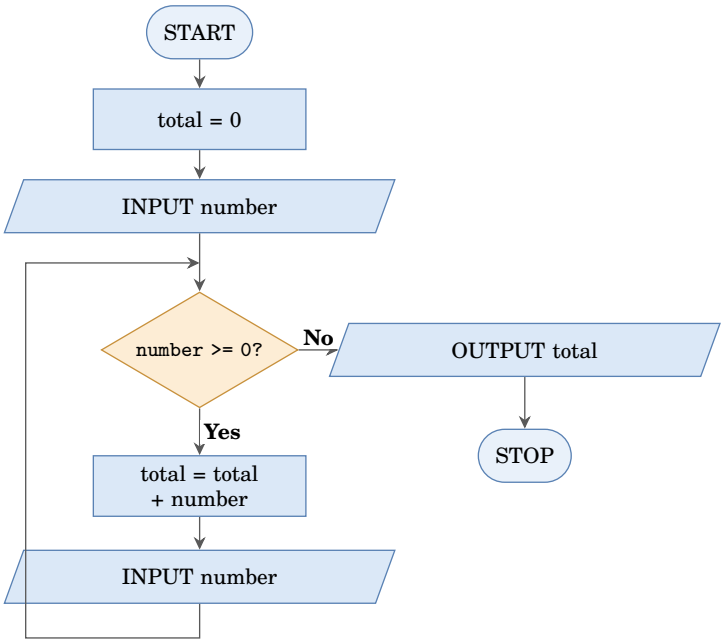
Symbol	Name	Use it for
START	Terminal	The start and the end. Every flowchart begins with START and finishes with STOP.
INPUT number	Input / output	Data coming in from the user, or going out to the screen.
total = 0	Process	A step that does something: a calculation, or giving a variable a value.
age > 17?	Decision	A question with two answers. It has two exits, labelled **Yes** and **No**.
show_menu()	Sub program	Running a sub program that is defined somewhere else.
→	Line	Joins the symbols. The arrow shows which step comes next.

2 A flowchart beside its program

The rule. Start at START and follow the arrows. At a decision, answer the question and leave by the exit with that label. A **loop** is an arrow that goes back up to a decision, so the steps below it run again.

Worked example: Sum of Positive Numbers, from your holiday sheet.

```
1 total = 0
2 number = int(input("Enter a number: "))
3 while number >= 0:
4     total = total + number
5     number = int(input("Enter a number: "))
6 print("The sum of positive numbers is:", total)
```



Shape	Line of the program
Process	1: `total = 0`
Input	2 and 5: `int(input(...))`
Decision	3: the `while` condition
Process	4: add to the total
Output	6: `print(...)`

The arrow from the second INPUT goes back up to the decision: that is the `while` loop. The decision comes **before** the steps it repeats, exactly as the `while` line comes before its indented lines.

Now try • Draw the flowchart for your **Even or Odd** holiday program. Use `MOD`, the exam's word for %: the decision is `number MOD 2 == 0`?

Check • One START, one STOP, one decision with both exits labelled, and an output on each exit: `Even` on the Yes side, `Odd` on the No side. Follow it with 7 and you reach `Odd`.

3 Trace tables

VOCAB **Trace table** (OCR 2.1.2) — a table used to follow an algorithm one line at a time, recording the value of each variable, to check what it does and to find logic errors

The rule. Give each variable its own column, plus a column for the condition and one for the output. Go through the program one line at a time, in the order the computer runs them. **When a line gives a variable a value, write that value in the variable's column on a new row.** When a line prints, write what it prints in the Output column.

Worked example. The Sum of Positive Numbers program above, with the inputs **5**, **12** and **−1**.

Line	number	number >= 0	total	Output
1			0	
2	5			
3		True		
4			5	
5	12			
3		True		
4			17	
5	−1			
3		False		
6				The sum of positive numbers is: 17

Lines 3, 4 and 5 repeat, once for each positive number. The loop ends when the condition is **False**, and only then does line 6 run. The last value in the total column is the value that gets printed.

Now try • Trace the same program with the inputs **8**, **0**, **3** and **−2**.

Check • The output is `The sum of positive numbers is: 11`. The condition column reads True three times, then False once: 0 is *0 or more*, so it goes round the loop.

Line	number	number >= 0	total	Output

4 The exam writes it in ERL

Exam trace tables are usually set on OCR Exam Reference Language (ERL). It is the same program: ERL closes the loop with `endwhile` where Python uses the indent, and joins text with +.

PYTHON (what you type)

```
total = 0
number = int(input("Enter a number: "))
while number >= 0:
    total = total + number
    number = int(input("Enter a number: "))
print("The sum of positive numbers is:", total)
```

OCR ERL (what the exam shows)

```
total = 0
number = int(input("Enter a number: "))
while number >= 0
    total = total + number
    number = int(input("Enter a number: "))
endwhile
print("The sum of positive numbers is: " + str(total))
```

A counting loop in ERL. `for i = 1 to 3` gives `i` the values 1, 2 and 3, **including** the 3. It is Python's `for i in range(1, 4):`, and `next i` closes the loop.

Now try • Trace this ERL program. It is your Multiplication Table holiday program for the number 7, cut to three lines of the table.

```
number = 7
for i = 1 to 3
    answer = number * i
    print(answer)
next i
```

Check • The Output column reads 7, 14, 21, and `i` takes the values 1, 2 and 3: the first three lines of the 7 times table.

Line	number	i	answer	Output

Stretch Guess the Number as a flowchart

Draw the flowchart for your **Guess the Number** holiday program. The `while` loop is a decision with the arrow coming back up to it, as in Section 2. Inside the loop, the `if`, `elif` and `else` become two decisions, one below the other.

Check • One START and one STOP, three decisions (`guess != secret?`, `guess > secret?`, `guess < secret?`), and every output inside the loop has an arrow back up to the first decision.

The exam answers, one line each

Asked for	Write
Flowchart	A diagram of an algorithm using standard symbols joined by arrows.
Decision symbol	A diamond, for a question with two exits labelled Yes and No.
Trace table	A table that records the value of each variable as an algorithm runs, line by line.
Why use a trace table	To check what an algorithm outputs, and to find logic errors.